

Gauge Group Structure Theorem, Direct Product Theorem, and the Three-Generation Proposition

— The Inevitable Emergence of $SU(3) \times SU(2) \times U(1)$ from Regime Boundaries and the Topological Constraint on the Number of Heavy Quarks(III)

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Abstract

Building on the empirical findings and mathematical theorems of the previous two papers, this paper completes the theoretical derivation of the gauge group structure of the Standard Model from the Factor Hierarchy Law. The meta-regime factor (color charge) satisfies three conditions: dividing particles into mutually exclusive regimes, having irreducible interaction effects, and modulating all executive factors — these conditions mathematically define exactly the generators of a discrete symmetry group. Given that color charge is a meta-regime factor with three mutually exclusive regime states (the latter being a known fact of the Standard Model), $SU(3)$ is the unique gauge group satisfying the conditions. The orthogonal independence of color charge and flavor quantum numbers (confirmed in Paper 15) implies that the two meta-regime factors act independently on distinct internal spaces, so their gauge groups must form a direct product. The $SU(3) \times SU(2) \times U(1)$ structure of the Standard Model is not an arbitrarily written assumption but an inevitable algebraic consequence of this rule. The topological constraint of the transition zone width, together with the experimental exclusion of a fourth generation of quarks by the LHC, jointly points to the number of heavy quarks being limited to four. The four regime centers (strange, charm, bottom, top) fill the mass range from approximately 1.5 GeV to approximately 172.6 GeV. The spacing from bottom to top (162.4 GeV) already far exceeds those from strange to charm (0.8 GeV) and from charm to bottom (7.17 GeV); the spacing of a fifth regime center would be even larger, and its transition zone width would cover a vast region above the electroweak unification scale. The two theorems and one proposition proved in this paper — the Gauge Group Structure Theorem (Theorem 4), the Direct Product Theorem (Theorem 5), and the Three-Generation Proposition (Proposition 6) — together constitute the demonstration of the inevitable emergence of the gauge group structure from regime boundaries. Gauge field theory is not the most fundamental theoretical framework — it is the expression of regime boundaries in the language of group theory. The Factor Hierarchy Law is an organizing principle deeper than gauge field theory. As the third paper of the five-paper series, this paper completes this crucial theoretical leap, providing the core theoretical pillar for the conceptual reinterpretations and system legislation of the subsequent two papers.

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Keywords: Factor Hierarchy Law; Testability Norms; gauge group structure; direct product theorem; three-generation proposition; SU(3); color charge; transition zone width

1. Introduction

1.1 From Regime Boundaries to Gauge Groups: The Empirical Foundation of the Previous Two Papers and the Unfinished Theoretical Task

Paper 14 (Tang, 2026n) used the Factor Hierarchy Law and Testability Norms to discover two regime-switching centers — the charm region (~ 2.29 GeV) and the bottom region (~ 9.46 – 10.02 GeV) — in the public dataset of the Particle Data Group (PDG) 2025. Color charge was confirmed as the meta-regime factor, and Mass was confirmed as a dual-identity factor. The algorithm locked the positions of Λ_c^+ , $Y(1S)$, and $Y(2S)$ with 10^{-4} GeV precision.

Paper 15 (Tang, 2026o) further discovered the strange region (~ 1.4 – 1.6 GeV) as the third regime center, revealed the precursor cascade — the fluctuation transformation point (T_2) of each regime center is precisely the precursor boundary of the next center. Cascade asymmetry was confirmed as a processual regime factor. Three mathematical theorems — the Transition Zone Width Theorem (Theorem 1), the Switching Type Dichotomy Theorem (Theorem 2), and the Switching Type Intrinsic Property Theorem (Theorem 3) — were formally proved. Theorem 1 established the rigorous equality between the inter-center transition zone width and the sum of the internal transition zone widths plus the gap. Theorem 2 proved that the intercept/slope ratio must be a dichotomy in finite samples. Theorem 3 proved that the switching type is an intrinsic property of the regime center.

However, the previous two papers left a fundamental theoretical task: **Why is the gauge group of the Standard Model precisely $SU(3) \times SU(2) \times U(1)$? Why are there exactly three generations of heavy quarks?** Traditional particle physics takes the gauge group structure as the starting point of the theory — it is assumed as given and not derived. The mission of this paper is to start from the statistical structure of regime boundaries and rigorously derive the inevitable form of the gauge group and the topological constraint on the number of heavy quarks. If successful, gauge field theory will be demoted from "the most

fundamental theoretical framework" to "the expression of regime boundaries in the language of group theory."

1.2 The Core Mission of This Paper

The core mission of this paper is to complete three theoretical derivations.

First, given that color charge is a meta-regime factor (confirmed in Paper 14) and has three mutually exclusive regime states (a known fact of the Standard Model), rigorously prove via the Cartan-Killing classification of Lie groups that $SU(3)$ is the unique gauge group satisfying the conditions — **the Gauge Group Structure Theorem (Theorem 4)**.

Second, starting from the orthogonal independence of color charge and flavor quantum numbers, rigorously prove that the gauge groups must form a direct product — **the Direct Product Theorem (Theorem 5)**.

Third, starting from the topological constraint of the transition zone width and the experimental exclusion of a fourth generation by the LHC, argue that the number of heavy quarks is limited to four within the currently experimentally accessible mass range — **the Three-Generation Proposition (Proposition 6)**.

The proofs of Theorem 4 and Theorem 5 require no additional physical assumptions — their premises have been confirmed in the more than 40 tests of the previous two papers, and the derivation logic is entirely based on the Cartan-Killing classification of Lie groups and the mathematical properties of structural break detection. The argumentative strength of Proposition 6 is weaker than that of the preceding five theorems — its conclusion depends on the empirical trend constituted by three data points and the independent experimental exclusion by the LHC; it is explicitly labeled as a "Proposition" rather than a "Theorem" in this paper.

1.3 The Position of This Paper in the Series

According to the PDCA division of labor of the five-paper series, this paper corresponds to the Check phase, covering the Why dimension of 5W2H. The division of labor with Papers 14 and 15 is shown in Table 1.

Table 1: Continuity Relationship with Papers 14 and 15

Dimension	Paper 14	Paper 15	Paper 16 (This Paper)
Regime centers	Charm, bottom	Strange	—

discovered			
Structural regularities	Activation operator stratification	Precursor cascade, transition zone width	—
Mathematical theorems	—	Theorems 1, 2, 3	Theorem 4, Theorem 5, Proposition 6
Theoretical derivation	—	—	Gauge group structure, direct product structure, number of generations
5W2H coverage	What, Who, When	How, How Much	Why

2. Axiomatic Foundation

The derivations in this paper rest upon the following axioms. These axioms have been cross-validated by at least two independent tools in the more than 40 tests of the previous two papers. Table 2 summarizes the empirical verification status of each axiom.

Table 2: Summary of Empirical Verification Status of the Axioms

Axiom	Core Content	Verification in Paper 14	Verification in Paper 15	Status
Axiom 1	Color charge is a meta-regime factor with a three-state structure	Interaction effect $p = 3.03 \times 10^{-3}$; after controlling for flavor quantum numbers $p = 3.81 \times 10^{-2}$	Strangeness effect confirmed within the color-charged regime	✓ Confirmed
Axiom 2	Color charge and flavor quantum numbers are orthogonally independent	Reverse proxy: p drops from 10^{-8} to 10^{-10}	Forward proxy confirmed	✓ Confirmed
Axiom 3	Inter-center transition zone width equals the sum of internal widths plus gap	—	Verified on three non-overlapping transition zones	✓ Confirmed

Axiom	Core Content	Verification in Paper 14	Verification in Paper 15	Status
Axiom 4	Precursor cascade: $T_2(C1) = \text{precursor}(C2)$	Charm $T_2 = \text{bottom precursor}$	Strange $T_2 \approx \text{charm } T_1$	✓ Confirmed
Axiom 5	Cartan-Killing classification of Lie groups	—	—	Established mathematical theorem

Axiom 1 (Three-state structure of the meta-regime factor): Color charge is the meta-regime factor.

It divides all particles into two mutually exclusive regimes — the color-charged regime and the color-neutral regime. Its interaction effect remains significant after controlling for all secondary regime factors (Paper 14, $p = 3.81 \times 10^{-2}$). Its modulating action covers all executive factors (Mass, Spin, Isospin). Within the color-charged regime, color charge manifests as three states — red, green, blue. **That color charge has three states is a known fact of the Standard Model, not a discovery of this paper. The contribution of this paper is to prove: given that color charge is a meta-regime factor (which is the discovery of Paper 14), its gauge group must necessarily be SU(3), rather than any other possible group structure.**

Axiom 2 (Orthogonal independence): Color charge and flavor quantum numbers are orthogonally independent meta-regime factors. When controlling for one, the modulating effect of the other not only does not vanish but becomes even more significant (Paper 15: after controlling for color charge, the Strangeness effect p drops from 10^{-8} to 10^{-10}). The two are not proxy relations but orthogonal parallel modulators.

Axiom 3 (Transition Zone Width Theorem): For any two adjacent regime centers, the inter-center transition zone width equals the sum of the two internal transition zone widths plus the gap between the centers (Paper 15, Theorem 1). On non-overlapping transition zones, this regularity has been verified on three transition zones.

Axiom 4 (Precursor cascade): The fluctuation transformation point (T_2) of each regime center is precisely the precursor boundary of the next center. Strange $T_2 \approx \text{Charm } T_1$ (deviation 0.0052 GeV), Charm $T_2 = \text{Bottom precursor}$ (deviation 0.0000 GeV). The masses of the four regime centers are approximately: strange ~ 1.5 GeV, charm ~ 2.29 GeV, bottom ~ 9.46 GeV, top ~ 172.6 GeV (CUSUM candidate).

Axiom 5 (Cartan-Killing classification of Lie groups): The compact simple Lie groups of rank 2 are only $SU(3)$, $SO(5)$, and G_2 (Cartan, 1894; Killing, 1888). This is an established mathematical theorem.

3. Theorem 4: Gauge Group Structure Theorem

3.1 Theorem Statement

Theorem 4 (Gauge Group Structure Theorem): Let a meta-regime factor F possess n mutually exclusive regime states, with its modulating action covering all executive factors and its interaction effect being irreducible. Then F necessarily generates a special unitary group $SU(n)$ acting on an n -dimensional complex vector space as its gauge group. Given that color charge is a meta-regime factor with $n = 3$ (the three-state structure being a known fact of the Standard Model), $SU(3)$ is the unique gauge group satisfying the conditions.

3.2 Proof

1. The meta-regime factor F divides particles into n mutually exclusive regime states. According to Axiom 1, color charge divides color-charged particles into three states — red, green, blue. These three states are indistinguishable in the strong interaction — they constitute the set of objects on which a symmetry group acts.
2. F 's modulating action covers all executive factors and is irreducible. This means the action of F is global — it acts on an n -dimensional internal space that is independent of spacetime coordinates.
3. A group G is required that can perform all regime-preserving transformations among the n states while keeping the regime boundaries unchanged. Mathematically, G must satisfy:
 - (a) G is a unitary group acting on an n -dimensional complex vector space, to preserve probability conservation;
 - (b) The group manifold of G is compact, to guarantee finite-dimensional representations;
 - (c) G is simple or contains a simple group, because F is irreducible — if G could be decomposed into a direct product, F would be reducible to two independent regime factors.

4. The group satisfying conditions (a)–(c) is the special unitary group $SU(n)$ (or contains $SU(n)$ as a subgroup). $SU(n)$ is the group of unitary transformations on an n -dimensional complex vector space with determinant 1. It is a compact simple Lie group whose group manifold has dimension (n^2-1) .
5. For color charge, $n = 3$. $SU(3)$ is a compact simple Lie group of rank 2, with 8 generators corresponding to the 8 gluons. According to Axiom 5, the compact simple groups of rank 2 are only $SU(3)$, $SO(5)$, and G_2 . The basic representation of $SO(5)$ is a 4-dimensional real spinor, which cannot produce a 3-dimensional complex representation; the basic representation of G_2 is 7-dimensional. Therefore, $SU(3)$ is the unique rank-2 compact simple group capable of producing a 3-dimensional complex representation.
6. Hence, $SU(3)$ is not assumed — given that color charge is a meta-regime factor with a three-state structure, $SU(3)$ is the unique gauge group satisfying the conditions.

Q.E.D.

3.3 Supplementary Discussion: Correspondence Between the Number of Regime Boundaries and the Number of Generators

Theorem 4 provides a path to derive $SU(3)$ from the three-state structure of the meta-regime factor. Additionally, the data from the previous two papers reveal a noteworthy correspondence: the three confirmed regime centers (strange, charm, bottom) each internally exhibit double-peak fission, producing a total of six internal regime boundaries. Adding the main external switching boundary of each center, as well as the hidden internal structure of the bottom region revealed by $W \times M$ and $W \times M^2$, the total number of regime boundaries in the strong-interaction sector falls between six and eight. The number of generators of $SU(3)$ is precisely eight.

It must be candidly stated that this correspondence is currently a numerical coincidence and has not yet been rigorously proved as a mathematical necessity. "The number of regime boundaries equals the number of generators" is a conjecture that awaits further argumentation; its establishment requires confirmation of the internal double-peak fission of the top region and more rigorous mathematical derivation. This conjecture will be further discussed in Paper 5 in conjunction with the formalization of the regime boundary algebra.

4. Theorem 5: Direct Product Theorem

4.1 Theorem Statement

Theorem 5 (Direct Product Theorem): Let F_1 and F_2 be two orthogonally independent meta-regime factors, possessing n_1 and n_2 mutually exclusive regime states, respectively. Then the complete gauge group of the system is the direct product $G_1 \times G_2$, where $G_1 = \text{SU}(n_1)$ and $G_2 = \text{SU}(n_2)$. Given that color charge ($n_1 = 3$) and weak isospin ($n_2 = 2$) are orthogonally independent in the Standard Model, the complete gauge group is $\text{SU}(3) \times \text{SU}(2) \times \text{U}(1)$, where the factor-level positioning of $\text{U}(1)$ awaits further analysis.

4.2 Proof

1. According to Axiom 2, color charge and flavor quantum numbers are orthogonally independent meta-regime factors. In the Standard Model, the change of flavor quantum numbers occurs through the weak interaction (propagated by the W boson). Therefore, the orthogonal independence of color charge and flavor quantum numbers implies that the regime boundaries of the strong interaction and the weak interaction are orthogonal.
2. F_1 and F_2 each independently generate their respective regime boundaries. Color charge generates the color-charged/color-neutral dichotomous regime, and weak isospin generates the left-handed/right-handed dichotomous regime. A particle can simultaneously be in a certain regime state of color charge and a certain regime state of weak isospin — the combination of the two regime boundaries is a Cartesian product.
3. Let G_1 be the gauge group generated by F_1 , and G_2 be the gauge group generated by F_2 . The two groups act independently on different internal spaces — color space and isospin space. Because F_1 and F_2 are orthogonally independent, the generators of G_1 commute with those of G_2 : $[T_i(G_1), T_j(G_2)] = 0$.
4. The joint action of two mutually commuting groups is described by the direct product $G_1 \times G_2$. The generators of the direct product are the union of the respective generators, and the group manifold of the direct product is the Cartesian product of the respective group manifolds.
5. According to Theorem 4, color charge ($n_1 = 3$) generates $\text{SU}(3)$. Weak isospin ($n_2 = 2$) generates $\text{SU}(2)$. The electromagnetic interaction is described by $\text{U}(1)$ — within the framework of the Factor Hierarchy Law, the physical status of $\text{U}(1)$ differs from that of $\text{SU}(3)$ and $\text{SU}(2)$: $\text{SU}(3)$ and $\text{SU}(2)$ are respectively

generated by color charge and weak isospin as meta-regime factors, whereas $U(1)$ describes hypercharge symmetry, whose number of regime states is 1, and which does not produce a mutually exclusive dichotomous regime. The factor-level positioning of $U(1)$ — whether it is a meta-regime factor or a derived symmetry — is a question to be further analyzed in Paper 4 in conjunction with the regime function of the Higgs mechanism.

6. Therefore, the direct product structure $SU(3) \times SU(2) \times U(1)$ is not an arbitrarily written assumption — it is the inevitable algebraic consequence of color charge and weak isospin being orthogonally independent meta-regime factors.

Q.E.D.

4.3 Supplementary Remarks: Distinction Between $U(1)_Y$ and $U(1)_{em}$

In the Standard Model, what forms a direct product with $SU(2)$ is $U(1)_Y$ (hypercharge symmetry), not $U(1)_{em}$ (electromagnetic symmetry). Electroweak breaking is the process $SU(2) \times U(1)_Y \rightarrow U(1)_{em}$; $U(1)_{em}$ is the residual Abelian symmetry after breaking, whose generator is a linear combination of the third generator of $SU(2)$ and the generator of $U(1)_Y$. Within the framework of the Factor Hierarchy Law, $U(1)_Y$ has a regime-state count of 1 and does not produce a mutually exclusive dichotomous regime — its factor-level positioning (whether it is a meta-regime factor or a derived symmetry) is a question to be further analyzed in Paper 4 in conjunction with the regime function of the Higgs mechanism. $U(1)_{em}$ is clearly not a meta-regime factor — it is the residual symmetry after electroweak regime breaking and does not generate an independent regime boundary.

Table 3: Open Questions Regarding the Factor-Level Positioning of $U(1)$

Symmetry	Number of Regime States	Produces Dichotomous Regime?	Factor-Level Positioning	Argumentative Path
$U(1)_Y$ (hypercharge)	1	No	To be analyzed (Paper 4)	In conjunction with the regime function of the Higgs mechanism
$U(1)_{em}$ (electromagnetic)	1	No	Not a meta-regime factor	Residual symmetry after electroweak breaking

5. Proposition 6: Three-Generation Proposition

5.1 Proposition Statement

Proposition 6 (Three-Generation Proposition): Within the currently experimentally accessible mass range, the number of heavy quarks is limited to four by the topological constraint of the transition zone width. The four regime centers (strange, charm, bottom, top) are the maximum number detectable in the current data. The number of generations is not arbitrary but is the inevitable consequence jointly indicated by the topological constraint of regime switching on the mass axis and the experimental facts from the LHC.

The argumentative strength of Proposition 6 is weaker than that of Theorem 4 and Theorem 5, and also weaker than that of Theorems 1 through 3 in Paper 15. Theorems 1 through 5 are rigorous mathematical derivations from axioms. The argumentation of Proposition 6 depends on the empirical trend constituted by three data points (transition zone width increasing with center spacing) and the independent experimental exclusion by the LHC; it is explicitly labeled as a "Proposition" rather than a "Theorem" in this paper. A complete mathematical proof requires more data on regime centers (such as confirmation of the internal double-peak fission of the top region) and tests from experiments at higher energy scales.

5.2 Argumentation

1. According to Axiom 4, the masses of the four regime centers are approximately: strange ~ 1.5 GeV, charm ~ 2.29 GeV, bottom ~ 9.46 GeV, top ~ 172.6 GeV (CUSUM candidate). The spacings between adjacent centers are: strange $\rightarrow$ charm ≈ 0.8 GeV, charm $\rightarrow$ bottom ≈ 7.17 GeV, bottom $\rightarrow$ top ≈ 162.4 GeV.
2. In the current data, the transition zone width increases as the center spacing increases: strange internal width 0.0852 GeV (detected by the Width operator), charm internal width 0.1865 GeV (Width operator), bottom internal width 3.517 GeV ($W \times M / W \times M^2$ activation operators). The three data points constitute an increasing trend — the larger the spacing, the wider the internal transition zone. It must be candidly noted that the bottom internal width is detected by activation operators, which differs from the detection mode of the strange internal width and charm internal width (both by the Width operator). Whether internal widths detected by different activation operators can be placed in the same trend is a question requiring further investigation.
3. Suppose a fifth regime center exists, with its mass lying above the top quark. Following the increasing trend of the bottom $\rightarrow$ top spacing of 162.4 GeV, the spacing of the fifth center would be even larger —

possibly reaching the order of 10^3 GeV or higher. At such a high mass scale, the transition zone width of the regime center would cover a vast mass range.

4. More importantly, within the currently experimentally accessible mass range (up to approximately 173 GeV, corresponding to the top quark), no additional regime-switching center has been detected. The multi-interval blank control scans in Papers 14 and 15 have confirmed: above the bottom region (11.0–50.0 GeV) no particles possess complete Mass+Spin+Isospin data.
5. The LHC has already excluded a fourth generation of quarks with masses below approximately 1 TeV (CMS Collaboration, 2019; ATLAS Collaboration, 2020). This independent experimental fact and the topological constraint of the transition zone width point to the same conclusion: within the current mass range, a fifth regime center cannot exist.
6. Therefore, within the currently experimentally accessible mass range, the number of heavy quarks is jointly limited to four by the topological constraint of the transition zone width and the experimental exclusion facts from the LHC.

Argumentation complete.

5.3 Positioning of Proposition 6 and Premises Awaiting Verification

The argumentation of Proposition 6 is mathematically less rigorous than the preceding five theorems. Its core depends on two conditions: first, the empirical trend that "transition zone width increases with center spacing" — currently only three data points exist (strange, charm, bottom), and the establishment of the trend requires data from more regime centers; second, the establishment of the top quark as the fourth regime center — currently the top region has only been detected by CUSUM and has not yet completed multi-tool cross-validation.

Once these premises are verified, Proposition 6 will be elevated from a "Proposition" to a "Theorem." The verification paths are clear: confirmation of the internal double-peak fission of the top region requires a larger sample of top quark data; the LHC's continued searches at higher energy scales will continually test whether a fifth regime center exists above the fourth spacing (bottom→top). Proposition 6 provides explicit testable predictions for these two experimental directions.

Table 4: Premises of the Three-Generation Proposition Awaiting Verification and Their Verification Paths

Premise Awaiting Verification	Current Evidence	Verification Path
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Premise Awaiting Verification	Current Evidence	Verification Path
Transition zone width increases with spacing	Three data points (strange, charm, bottom)	Confirmation of internal double-peak fission of the top region
Top quark is the fourth regime center	CUSUM detection (candidate)	Multi-tool cross-validation with a larger sample
Fifth regime center does not exist	LHC exclusion < 1 TeV	LHC searches at higher energy scales

6. Discussion

6.1 The Demotion of Gauge Field Theory

Theorem 4, Theorem 5, and Proposition 6 of this paper jointly accomplish a fundamental theoretical transformation: gauge field theory is demoted from "the most fundamental theoretical framework" to "the expression of regime boundaries in the language of group theory."

When gauge field theory was proposed by Yang and Mills (Yang & Mills, 1954) in 1954, it was entirely reasoning based on mathematical self-consistency, without any experimental evidence. When the SU(2) gauge group was proposed, physicists were not even sure whether the weak interaction was propagated by gauge bosons. It was not until the 1970s that the SU(3)×SU(2)×U(1) structure of the Standard Model was gradually verified by experiment.

By contrast, the derivations in this paper have a stronger empirical foundation. The existence of the meta-regime factor (Axiom 1) was confirmed by the interaction effect tests in Paper 14. Orthogonal independence (Axiom 2) was confirmed in Paper 15. The Transition Zone Width Theorem (Axiom 3) has been verified on three non-overlapping transition zones. The precursor cascade (Axiom 4) has been precision-quantitatively verified on three cascade pairs. These more than 40 tests constitute a stronger empirical foundation than what gauge field theory had at the time of its proposal.

Gauge field theory is not overturned — it is demoted. The mathematical structure of SU(3)×SU(2)×U(1) remains correct, but it is no longer "the most fundamental." The Factor Hierarchy Law is the deeper organizing principle, and gauge field theory is its specific expression in the continuous symmetry limit. This relationship is structurally completely isomorphic to the argument in Paper 13 (Tang, 2026m) that "universality is a mathematical convergence product in the thermodynamic limit, while regime boundaries are the fundamental quantities in finite systems."

6.2 Connection with the 2D Ising Information Duality

Paper 11 (Tang, 2026k) proved, on the Onsager exact solution of the 2D Ising model, the precise information-theoretic duality between the Factor Hierarchy Law and the Ehrenfest classification of phase transitions — the Chow test attains its global unique maximum at the exact T_c , with the deviation analytically proved to be strictly zero. That proof is the mathematical cornerstone of the particle physics series of derivations: it establishes the information equivalence between the Chow test and physical phase transitions.

The Gauge Group Structure Theorem of this paper extends this information duality from phase transition classification to gauge symmetry — the same information equivalence exists between the statistical structure of meta-regime factors and the mathematical structure of gauge groups. The Chow test detects a structural break at the color charge regime boundary, and the position of this break exactly coincides with the action boundary of the $SU(3)$ gauge group. The generalized form of the information duality can be stated as: **Any regime boundary generated by a meta-regime factor necessarily corresponds, in group theory, to a gauge group.**

It must be prudently noted that the converse proposition — "any generator of a gauge group necessarily corresponds, in statistics, to a structural breakpoint" — is currently still a conjecture. Its validity requires two conditions: the symmetry corresponding to that generator must produce a detectable regime boundary in the data, and that regime boundary must fall within the detection range of current structural break tools. The rigorous proof of this conjecture is one of the core tasks of the formalization of the regime boundary algebra in Paper 5.

6.3 The Physical Picture of the Direct Product Structure

The $SU(3) \times SU(2) \times U(1)$ direct product structure of the Standard Model acquires an entirely new physical interpretation within the framework of the Factor Hierarchy Law. It is not a simple juxtaposition of "the mathematical descriptions of three interactions" — it is the inevitable algebraic consequence of two orthogonally independent meta-regime factors.

Color charge determines whether a particle participates in the strong interaction. Weak isospin determines how a particle participates in the weak interaction. The two meta-regime factors are orthogonally independent — a particle can simultaneously carry color charge and weak isospin, with the two quantum numbers transforming independently in different internal spaces. This orthogonal independence necessarily

manifests in group theory as a direct product — because the generators of the two groups commute, they act on different internal spaces.

This is completely consistent with the standard formulation in quantum field theory that "the gauge group is a direct product," but provides a deeper explanation: it is not that "the gauge group is a direct product because three interactions are observed experimentally," but that "because two orthogonally independent meta-regime factors exist, the gauge group must be a direct product." **The causal direction has been reversed.**

6.4 The Physical Picture of the Three-Generation Proposition

"Why exactly three generations?" is one of the deepest questions in particle physics. The answer provided by Proposition 6 of this paper is: **The topological constraint of the transition zone width and the experimental exclusion facts from the LHC jointly point to three generations.**

This answer requires candid acknowledgment of its argumentative boundaries. The empirical trend of transition zone width increasing with center spacing currently has only three data points (strange, charm, bottom). The internal double-peak fission of the top region has not yet been confirmed, and the trend beyond the bottom→top spacing of 162.4 GeV is currently still an extrapolation. The experimental fact that the LHC excludes a fourth generation of quarks is independent, but its exclusion range (below approximately 1 TeV) does not completely cover all mass intervals where a fifth regime center could exist.

The complete verification of Proposition 6 requires two conditions: first, confirmation of the internal double-peak fission of the top region, which will provide the fourth data point for the regularity of transition zone width increasing with spacing; second, continued searches by the LHC at higher energy scales, which will provide more stringent experimental constraints on the exclusion of a fifth regime center. Proposition 6 provides explicit testable predictions for these two experimental directions.

7. Conclusion

Building on the empirical findings and three mathematical theorems of the previous two papers, this paper has completed the core theoretical derivation of the gauge group structure of the Standard Model from the Factor Hierarchy Law.

Theorem 4 (Gauge Group Structure Theorem): The mutually exclusive regime-state structure of a meta-regime factor necessarily generates the corresponding gauge group. Given that color charge is a

meta-regime factor with a three-state structure, $SU(3)$ is the unique gauge group satisfying the conditions. The Cartan-Killing classification of Lie groups excludes $SO(5)$ and G_2 as candidate groups.

Theorem 5 (Direct Product Theorem): Orthogonally independent meta-regime factors necessarily lead to a direct product of gauge groups. The orthogonal independence of color charge and flavor quantum numbers (confirmed in Papers 14 and 15) means that the direct product structure of $SU(3)$ and $SU(2) \times U(1)$ is not an assumption but an algebraic necessity. The factor-level positioning of $U(1)$ is left for further analysis in Paper 4.

Proposition 6 (Three-Generation Proposition): The topological constraint of the transition zone width and the experimental exclusion of a fourth generation of quarks by the LHC jointly point to the number of heavy quarks being limited to four within the currently experimentally accessible mass range. The argumentative strength of Proposition 6 is weaker than that of the preceding five theorems — its core depends on the empirical trend constituted by three data points and the independent experimental exclusion; a complete mathematical proof requires confirmation of the internal double-peak fission of the top region and tests from experiments at higher energy scales.

These two theorems and one proposition jointly accomplish a fundamental theoretical transformation: **Gauge field theory is demoted from "the most fundamental theoretical framework" to "the expression of regime boundaries in the language of group theory."** The Factor Hierarchy Law is an organizing principle deeper than gauge field theory. The empirical foundation of this conclusion is stronger than what gauge field theory had at the time of its proposal — more than 40 tests, six structural break tools, 11 activation operators, a 2569-fold activation effect, and 10^{-4} GeV precision deviations.

This paper is the third of the five-paper series. The subsequent fourth paper will use regime boundary theory to reinterpret the statistical origin of CP violation, the regime function of the Higgs mechanism, and the statistical correspondence of virtual intermediate states. The fifth paper will complete the formalization of the five-layer axiomatization structure and the final establishment of the testability program.

Data Availability Statement

This paper is a theoretical derivation paper and involves no new data analysis. The empirical foundations of all axioms and premises have been reported in detail in Paper 14 (Tang, 2026n) and Paper 15 (Tang, 2026o). The data are completely open, and any researcher can independently reproduce all empirical results.

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